확률·랜덤프로세스 기말대비 선별 5문항 — Lecture 01~06 중간범위 종합(기초확률·이산/연속RV) (★=핵심)

① 중간범위 복습 — 조건부·베이즈·PMF/PDF 기본기 ② 분포 보이면 대표공식(이항 np, 지수 1/λ, 정규 표준화) 먼저 ③ 채점: origin 풀이 / /solve-problem

떠올릴 개념

- 조건부확률 P(A|B)=P(A∩B)/P(B); 전확률정리 P(B)=Σ P(B|Ai)P(Ai)
- 베이즈 P(A|B)=P(B|A)P(A)/P(B); 독립 ⇔ P(A∩B)=P(A)P(B)
- 이산 기댓값 E[X]=Σ x·P(x), Var=E[X^2]-(E[X])^2
- 대표 이산분포: 이항 E=np Var=np(1-p) · 기하 E=1/p · 균등(이산)
- 무기억성(기하/지수) P(X>s+t | X>s)=P(X>t)
- 연속 E[X]=∫ x f(x) dx, F(x)=∫ f, f=F'; ∫ f(x) dx=1로 상수 결정
- 지수분포 f=λe^(-λx) (x≥0); 정규 표준화 Z=(X-μ)/σ → Φ/Q함수
- 모든 PMF/PDF는 명시 구간 밖에서 otherwise = 0

문항 목차 (총 5문항)

1.4.4 Phonesmart is having a sale on Bananas. If you buy one Banana at full price, you get a second at half price. When couples come in to buy a pair of phones, sales of Apricots and Bananas are equally likely. Moreover, given that the first phone sold is a Banana, the second phone is twice as likely to be a Banana rather than an Apricot. What is the probability that a couple buys a pair of Bananas?

— 풀이 —

3.4.5 At the One Top Pizza Shop, a pizza sold has mushrooms with probability $p = 2/3$. On a day in which 100 pizzas are sold, let N equal the number of pizzas sold before the first pizza with mushrooms is sold. What is the PMF of N? What is the CDF of N?

— 풀이 —

5.3.5 Random variables N and K have the joint PMF

$$P_{N,K}(n,k) = \begin{cases} \frac{(1-p)^{n-1}p}{n} & \substack{k=1,\ldots,n;\\ n=1,2,\ldots} \\ 0 & \text{otherwise.} \end{cases}$$

Find the marginal PMFs $P_N(n)$ and $P_K(k)$.

— 풀이 —

Problem 2

Let X be a continuous random variable with the following PDF

$$f_X(x) = \begin{cases} ce^{-4x} & x \geq 0 \\ 0 & otherwise \end{cases}$$

where c is a positive constant.

a. Find c.
b. Find the CDF of X, $F_X(x)$.
c. Find $P(2 < X < 5)$.
d. Find EX.

— 풀이 —

4.6.3● Find each probability.

(a) V is a Gaussian $(\mu = 0, \sigma = 2)$ random variable. Find $P[V > 4]$.

(b) W is a Gaussian $(\mu = 2, \sigma = 5)$ random variable. What is $P[W \leq 2]$?

(c) For a Gaussian $(\mu, \sigma = 2)$ random variable X, find $P[X \leq \mu + 1]$.

(d) Y is a Gaussian $(\mu = 50, \sigma = 10)$ random variable. Calculate $P[Y > 65]$.

— 풀이 —